SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA0402**

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01 **Total Marks:100**

Annexure A

CASE STUDY

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jannathul Firdaus. N

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 1: E-Commerce Customer Analysis
Scenario:
An e-commerce company has collected customer transaction data, including purchase amount, number of orders, browsing time, and customer ratings. The company wants to understand customer purchasing behavior and improve its marketing strategies.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize customer transaction data?
B. Discuss how unusually high or low purchase amounts can affect the analysis and describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether premium members spend significantly more than regular customers? Formulate the hypotheses and explain the analysis process.

Case Study 2: Renewable Energy Production Analysis
Scenario:
A solar power company has collected data from multiple solar plants, including daily energy generation, sunlight hours, panel temperature, and system efficiency. The company wants to evaluate plant performance and verify whether the average energy production meets the expected target.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize and evaluate the energy production dataset.
B. Discuss the impact of abnormal energy generation values on the analysis and describe methods for detecting and handling outliers.
C. How can confidence intervals and hypothesis testing be used to determine whether the average daily energy generation differs significantly from the expected target value?

Case Study 3: Hospital Patient Recovery Analysis
Scenario:
A hospital has collected patient recovery data, including age, treatment cost, number of hospital days, recovery time, and patient satisfaction score. The hospital wants to analyze treatment effectiveness and improve patient care quality.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize patient recovery time and treatment cost?
B. Discuss how unusually high hospital stay duration or treatment cost values may affect the analysis. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether patients receiving a new treatment recover faster than patients receiving the existing treatment? Formulate the hypotheses and explain the analysis process.

Case Study 4: Student Academic Performance Analysis
Scenario:
A university has collected student performance data, including attendance percentage,

assignment marks, internal test scores, study hours, and final exam marks. The university wants to identify factors affecting student success.

Questions:

A. Explain how descriptive statistics and distribution analysis can be used to summarize student marks and attendance data.

B. Discuss how extremely low or high marks can affect the interpretation of academic performance. Describe suitable methods for detecting and handling outliers.

C. How can hypothesis testing be used to check whether students with attendance above 75% score significantly higher than students with attendance below 75%? Formulate the hypotheses and explain the steps.

Case Study 5: Banking Loan Approval Analysis

Scenario:

A bank has collected customer loan application data, including income, credit score, loan amount, repayment history, employment status, and loan approval status. The bank wants to understand customer risk and improve loan approval decisions.

Questions:

A. How can descriptive statistics be used to summarize customer income, credit scores, and loan amounts?

B. Discuss how unusually high loan amounts or very low credit scores can influence the analysis. Explain methods for identifying such outliers.

C. How can hypothesis testing be used to determine whether customers with higher credit scores are more likely to receive loan approval than customers with lower credit scores? Formulate the hypotheses and describe the analysis process.

Case Study 6: Online Learning Platform Analysis

Scenario:

An online learning platform has collected learner activity data, including course completion percentage, quiz scores, video watch time, login frequency, and feedback ratings. The platform wants to improve course engagement and learner success.

Questions:

A. Explain how descriptive statistics can be used to summarize quiz scores, watch time, and course completion rates.

B. Discuss how unusually high or low learning activity values may affect the analysis. What techniques can be used to detect outliers?

C. How can hypothesis testing be used to determine whether learners who watch more than 70% of the course videos score significantly higher in quizzes than other learners? Formulate the hypotheses and explain the testing procedure.

Case Study 7: Retail Sales Performance Analysis

Scenario:

A retail chain has collected sales data from different branches, including daily sales amount, number of customers, discount percentage, product category, and customer feedback score. The company wants to evaluate branch performance and improve sales planning.

Questions:

A. How can descriptive statistics and distribution analysis be used to summarize daily sales and customer count?

B. Discuss how abnormal sales values, such as festival-season sales or very low sales days, can affect the analysis. Explain suitable outlier detection methods.

C. How can confidence intervals and hypothesis testing be used to determine whether the average sales of urban branches are significantly higher than rural branches?

Case Study 8: Food Delivery Service Analysis
Scenario:
A food delivery company has collected order data, including delivery time, order value, customer rating, distance travelled, and delivery partner experience. The company wants to reduce delivery delays and improve customer satisfaction.
Questions:
A. Explain how descriptive statistics such as mean, median, standard deviation, and quartiles can be used to analyze delivery time and customer ratings.
B. Discuss how unusually long delivery times can affect customer satisfaction analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether experienced delivery partners complete deliveries faster than new delivery partners? Formulate the hypotheses and explain the analysis process.

Case Study 9: Manufacturing Defect Analysis
Scenario:
A manufacturing company has collected production data, including number of units produced, defect count, machine temperature, production time, and operator shift. The company wants to reduce defects and improve product quality.
Questions:
A. How can descriptive statistics be used to summarize defect counts, production time, and machine temperature?
B. Discuss how abnormal defect values may affect the quality analysis. What methods can be used to detect outliers in production data?
C. How can hypothesis testing be used to determine whether the defect rate differs significantly between day shift and night shift production? Formulate the hypotheses and explain the analysis process.

Case Study 10: Mobile Network Performance Analysis
Scenario:
A telecom company has collected mobile network data, including signal strength, internet speed, call drop rate, number of users, and customer complaints from different locations. The company wants to improve network quality.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize internet speed, signal strength, and call drop rate.
B. Discuss how extremely low internet speed or high call drop rate values can affect the analysis. Describe suitable outlier detection and handling methods.
C. How can confidence intervals and hypothesis testing be used to determine whether the average internet speed in urban areas is significantly different from rural areas?

Case Study 11: Public Transportation Usage Analysis
Scenario:
A city transport department has collected bus usage data, including passenger count, ticket revenue, waiting time, travel duration, and route distance. The department wants to improve transport scheduling and passenger service.

Questions:
A. How can descriptive statistics be used to summarize passenger count, waiting time, and travel duration?
B. Discuss how unusually high passenger counts during peak hours or unusually long waiting times can affect the analysis. Explain methods for detecting outliers.
C. How can hypothesis testing be used to determine whether average waiting time during peak hours is significantly higher than during non-peak hours? Formulate the hypotheses and explain the testing process.

Case Study 12: Agriculture Crop Yield Analysis
Scenario:
An agriculture department has collected crop production data from different farms, including rainfall, fertilizer usage, soil moisture, temperature, and crop yield. The department wants to analyze factors affecting crop productivity.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize crop yield, rainfall, and fertilizer usage.
B. Discuss how abnormal crop yield values caused by drought, pests, or excessive rainfall can affect the analysis. Describe suitable outlier detection methods.
C. How can confidence intervals and hypothesis testing be used to determine whether farms using organic fertilizer have significantly different crop yield compared with farms using chemical fertilizer?

Case Study 13: Employee Productivity Analysis
Scenario:
A software company has collected employee productivity data, including working hours, number of completed tasks, project deadlines, experience level, and performance rating. The company wants to improve employee performance and project planning.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize employee productivity data?
B. Discuss how unusually high or low task completion values can affect productivity analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether experienced employees complete significantly more tasks than newly joined employees? Formulate the hypotheses and explain the analysis process.

Case Study 14: Automobile Fuel Efficiency Analysis
Scenario:
An automobile company has collected vehicle performance data, including fuel consumption, engine capacity, vehicle weight, mileage, and maintenance cost. The company wants to analyze fuel efficiency and improve vehicle design.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize mileage and fuel consumption data.
B. Discuss how abnormal mileage values can affect the analysis. What methods can be used to detect and handle outliers?
C. How can confidence intervals and hypothesis testing be used to determine whether hybrid cars have significantly better mileage than petrol cars?

Case Study 15: Social Media Engagement Analysis
Scenario:
A digital marketing agency has collected social media data, including number of likes, shares, comments, followers, posting time, and engagement rate. The agency wants to improve content strategy and audience reach.
Questions:
A. How can descriptive statistics be used to summarize likes, shares, comments, and engagement rate?
B. Discuss how viral posts with extremely high engagement values may affect the analysis. Describe suitable outlier detection methods.
C. How can hypothesis testing be used to determine whether video posts receive significantly higher engagement than image posts? Formulate the hypotheses and explain the testing procedure.

Case Study 16: Airline Passenger Satisfaction Analysis
Scenario:
An airline company has collected passenger feedback data, including ticket price, flight delay time, baggage waiting time, seat comfort rating, and overall satisfaction score. The airline wants to improve service quality.
Questions:
A. Explain how descriptive statistics can be used to summarize delay time, baggage waiting time, and passenger satisfaction scores.
B. Discuss how unusually long flight delays can affect the analysis of passenger satisfaction. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights?

Case Study 17: Smart City Traffic Analysis
Scenario:
A smart city traffic department has collected traffic data, including vehicle count, average speed, signal waiting time, accident count, and pollution level from different roads. The department wants to reduce congestion and improve road safety.
Questions:
A. How can descriptive statistics and distribution analysis be used to summarize vehicle count, waiting time, and average speed?
B. Discuss how abnormal traffic values during accidents, festivals, or roadblocks can affect the analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether average waiting time is significantly higher during peak hours than during non-peak hours?

Case Study 18: Hotel Booking and Customer Review Analysis
Scenario:
A hotel chain has collected booking data, including room price, length of stay, customer rating, booking cancellation status, and service feedback score. The hotel wants to improve customer satisfaction and pricing strategy.

Questions:
A. How can descriptive statistics be used to summarize room price, length of stay, and customer ratings?
B. Discuss how unusually high room prices or extremely low customer ratings can affect the analysis. Explain suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether customers who stay more than three days give significantly different ratings compared with customers who stay for one or two days?

Case Study 19: Fitness App User Health Analysis
Scenario:
A fitness app has collected user health data, including daily steps, calories burned, workout duration, heart rate, sleep hours, and fitness goal completion. The app wants to study user activity patterns and improve health recommendations.
Questions:
A. Explain how descriptive statistics can be used to summarize daily steps, calories burned, workout duration, and sleep hours.
B. Discuss how unusually high workout duration or abnormal heart rate values can affect the health analysis. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether users who sleep at least seven hours per day burn significantly more calories than users who sleep less than seven hours?

Case Study 20: Online Examination Performance Analysis
Scenario:
An educational institution has collected online exam data, including exam score, time taken to complete the test, number of attempts, internet interruption count, and student feedback. The institution wants to evaluate student performance and improve the online assessment system.
Questions:
A. How can descriptive statistics and distribution analysis be used to summarize exam scores, completion time, and number of attempts?
B. Discuss how unusually short completion time or extremely high number of attempts can affect exam performance analysis. Describe suitable outlier detection methods.
C. How can hypothesis testing be used to determine whether students with fewer internet interruptions score significantly higher than students with frequent interruptions?

Case Study 2: Renewable Energy Production Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help summarize and understand the daily energy generation data collected from different solar plants.

1. **Mean:** The mean calculates the average daily energy production of a solar plant. It helps determine the overall production level and compare the performance of different plants.
2. **Median:** The median represents the middle value of energy generation when the observations are arranged in ascending order. It provides a reliable measure when extreme production values are present.
3. **Standard deviation:** It measures the variation in daily energy production. A low standard deviation indicates consistent production, while a high value indicates fluctuations caused by changing sunlight, temperature, or system performance.
4. **Quartiles and interquartile range:** Quartiles divide the data into four equal parts. The interquartile range (IQR), calculated as Q3 − Q1, measures the spread of the middle 50% of observations.
5. **Distribution analysis:** Histograms and box plots help identify skewness, variation, and unusual energy generation values. Scatter plots can reveal relationships between sunlight hours, panel temperature, and energy output.

Conclusion: These statistical measures help identify high-performing solar plants, understand production variability, and improve energy forecasting and maintenance planning.

B. Impact and Detection of Outliers

Abnormal energy generation values can significantly affect the analysis. Extremely high readings may result from measurement errors or incorrect data recording, while unusually low readings may indicate equipment failure, shading, cloudy weather, or system downtime.
Suitable methods for detecting outliers include:

- **IQR method:** Calculate Q1, Q3, and IQR. Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are flagged as potential outliers.
- **Z-score method:** Standardize observations using their mean and standard deviation. Values with an absolute Z-score greater than 3 are commonly investigated.
- **Box plots:** Visually identify observations outside the expected range.
- **Time-series analysis:** Compare daily generation with historical values and expected production under similar sunlight conditions.

Outliers should not be removed automatically. The company should verify sensor records, equipment logs, weather conditions, and maintenance reports. Genuine low-production events should be retained because they provide useful information about plant performance. Confirmed measurement errors may be corrected or excluded, while robust statistics can be used when extreme values remain valid.

C. Confidence Intervals and Hypothesis Testing

Confidence intervals and hypothesis testing can determine whether the average daily energy production differs significantly from the expected target.
Let μ represent the true mean daily energy generation and T represent the expected target.

Hypotheses:

- Null hypothesis (H_0): $\mu = T$
- Alternative hypothesis (H_1): $\mu \neq T$

A two-tailed one-sample t-test is appropriate when the population standard deviation is unknown.

Procedure:

1. Collect daily energy generation data from the solar plant over a representative period.
2. Calculate the sample mean, sample standard deviation, and sample size.
3. Construct a 95% confidence interval:

Sample mean ± t-critical × (sample standard deviation / $\sqrt{n}$).

4. Perform a one-sample t-test using a significance level of 0.05.
5. Reject H_0 if the p-value is less than 0.05. Otherwise, fail to reject H_0.

If the confidence interval excludes the target value, the results indicate a statistically significant difference at the corresponding significance level.

Solution: The company should use these results to identify underperforming plants, investigate production losses, and plan preventive maintenance. Statistical significance should also be considered alongside the practical size of the production difference.

Case Study 4: Student Academic Performance Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help the university understand student marks, attendance percentages, assignment scores, internal test results, study hours, and final examination performance.

1. **Mean:** Calculates the average marks or attendance percentage of students. It provides an overall measure of academic performance.
2. **Median:** Identifies the middle score and is useful when a few students obtain extremely high or low marks.
3. **Standard deviation:** Measures the variation in marks. A low value indicates that students perform similarly, while a high value indicates considerable differences in performance.
4. **Quartiles:** Divide the marks into four equal groups and help identify the performance levels of weaker, average, and stronger students.
5. **Distribution analysis:** Histograms show whether marks are normally distributed, negatively skewed, or positively skewed. Box plots help compare the spread of marks across subjects or student groups.
6. **Relationship analysis:** Scatter plots can examine whether study hours and attendance are associated with higher final examination marks.

Conclusion: The university can use these results to identify learning difficulties, evaluate attendance patterns, and develop targeted academic support programmes.

B. Impact and Detection of Outliers

Extremely high or low marks may distort the average and produce misleading conclusions about student performance. A few exceptionally low scores can reduce the mean, while unusually high scores can make overall performance appear better than it is.

Outliers may arise from data entry errors, missed examinations, illness, special circumstances, or genuinely exceptional performance.

Suitable detection methods include:

- **IQR method:** Flag marks below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.
- **Z-score method:** Investigate observations with absolute Z-scores greater than 3, where appropriate.
- **Box plots:** Identify unusual marks visually.
- **Data validation:** Check whether marks fall within the permitted scoring range and verify the original records.

The university should verify flagged observations before taking action. Incorrect entries should be corrected, while genuine low marks should be retained. If unusual observations substantially influence the mean, the median and trimmed mean may be reported as supplementary measures.

Solution: A transparent outlier-handling process ensures that academic decisions are based on accurate information rather than automatically excluding students with unusual scores.

C. Hypothesis Testing for Attendance and Marks

The university wants to determine whether students with attendance above 75% score significantly higher than students with attendance below 75%.

Let μ_1 be the mean final examination mark of students with attendance above 75%, and μ_2 be the mean mark of students with attendance below 75%.

Hypotheses:

- $H_0: \mu_1 \leq \mu_2$
- $H_1: \mu_1 > \mu_2$

A one-tailed independent-samples t-test, preferably Welch's t-test when group variances may differ, can be used.

Testing procedure:

1. Divide students into two groups based on the 75% attendance threshold.
2. Calculate the mean, standard deviation, and sample size for each group.
3. Check data quality, independence, and the suitability of the t-test. For small samples, examine the distributions and consider an appropriate alternative if assumptions are not met.
4. Conduct the one-tailed test at a 5% significance level.
5. Reject H_0 if the p-value is below 0.05.

Solution: If the results are statistically significant, the university has evidence that the higher-attendance group obtains higher average marks. However, this does not prove that attendance alone causes better performance because study habits, prior knowledge, and other factors may also influence marks.

The university should combine the findings with student support initiatives, attendance monitoring, and academic counselling.

Case Study 6: Online Learning Platform Analysis

A. Descriptive Statistics for Learner Activity

Descriptive statistics help the platform evaluate quiz scores, video watch time, login frequency, feedback ratings, and course completion percentages.

- **Mean:** Measures the average quiz score, video watch time, or course completion rate.
- **Median:** Identifies the middle observation and reduces the influence of learners with unusually high or low activity.
- **Standard deviation:** Shows whether learner activity is consistent or varies substantially across users.
- **Quartiles:** Divide learner activity into four groups and help identify learners with low engagement or high completion rates.
- **Histograms:** Show the distribution of quiz scores and video watch time.
- **Box plots:** Highlight potential outliers and compare engagement across courses.
- **Scatter plots:** Reveal whether video watch time is associated with quiz performance or course completion.

Conclusion: These measures help identify engagement patterns, assess course effectiveness, and design personalized learning interventions.

B. Impact and Detection of Outliers

Unusually high or low learning activity values can distort the analysis. Extremely high video watch time may result from videos being left running in the background, repeated playback, or tracking errors. Very low watch time may indicate genuine disengagement, technical problems, or incomplete tracking.

Suitable methods include:

1. **IQR method:** Identify observations outside $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.
2. **Z-score method:** Flag observations with unusually large standardized deviations where the distribution makes this method suitable.
3. **Box plots and histograms:** Identify extreme values and skewed distributions.
4. **Activity validation:** Compare watch-time records with video duration, login logs, and playback events.

The platform should validate flagged records before removing them. Invalid tracking records may be corrected or excluded, while genuine high-engagement learners should remain in the dataset. Robust measures such as the median can reduce the influence of extreme values.

Solution: Accurate outlier handling improves the reliability of learner engagement analysis and prevents incorrect conclusions about course participation.

C. Hypothesis Testing for Video Watch Time and Quiz Scores

The platform wants to determine whether learners who watch more than 70% of course videos achieve higher quiz scores than other learners.

Let μ_1 represent the mean quiz score of learners who watch more than 70% of the videos and μ_2 represent the mean score of learners who watch 70% or less.

Hypotheses:

- $H_0: \mu_1 \leq \mu_2$

- $H_1: \mu_1 > \mu_2$

An independent-samples t-test, preferably Welch's t-test, can compare the two groups.

Procedure:

1. Calculate each learner's percentage of course videos watched.
2. Divide learners into the two groups using the 70% threshold.
3. Calculate the mean quiz score and standard deviation for each group.
4. Check independence, data quality, and distributional assumptions.
5. Conduct a one-tailed t-test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.

Solution: A significant result suggests that learners who watch more than 70% of course videos achieve higher average quiz scores. The platform can use this evidence to encourage video completion, send progress reminders, and improve course design.

However, the result establishes an association rather than causation. Learner motivation, prior knowledge, and study time may also affect quiz scores.

Case Study 8: Food Delivery Service Analysis

A. Descriptive Statistics for Delivery Time and Customer Ratings

Descriptive statistics help the company evaluate delivery efficiency and customer satisfaction.

1. **Mean delivery time:** Measures the average time required to deliver orders and helps assess overall service speed.
2. **Median delivery time:** Represents the middle delivery time and is particularly useful when a few deliveries experience severe delays.
3. **Standard deviation:** Measures the consistency of delivery times. A high value indicates unpredictable service.
4. **Quartiles and IQR:** Show the distribution of delivery times and identify the range containing the middle 50% of orders.
5. **Customer rating statistics:** The mean and median summarize customer feedback, while the distribution of ratings shows the proportion of satisfied and dissatisfied customers.
6. **Distribution analysis:** Histograms and box plots reveal skewness and unusually long delivery times.
7. **Relationship analysis:** Scatter plots can examine how travel distance and delivery partner experience relate to delivery duration.

Conclusion: These measures help identify inefficient delivery routes, estimate expected delivery times, and improve customer satisfaction.

B. Impact and Detection of Unusually Long Delivery Times

Unusually long delivery times can increase the mean delivery duration and make the service appear less efficient than it usually is. Severe delays may also be associated with lower customer ratings and more complaints.

Possible causes include traffic congestion, adverse weather, incorrect addresses, restaurant preparation delays, and delivery partner shortages.

Suitable detection methods include:

- **IQR method:** Flag delivery times above Q3 + 1.5 × IQR as potential outliers.
- **Z-score method:** Investigate observations with unusually high standardized values when appropriate.
- **Box plots:** Display extreme delivery times visually.
- **Time and route comparisons:** Compare delivery times with historical values for similar distances, locations, and time periods.

The company should investigate flagged orders using GPS records, order timestamps, restaurant preparation data, and traffic information. Genuine delays should not be removed because they reveal operational problems. Confirmed data errors should be corrected.

Solution: The company should use validated delay data to improve route allocation, adjust estimated delivery times, and introduce additional resources during peak hours.

C. Hypothesis Testing for Experienced and New Delivery Partners

The company wants to determine whether experienced delivery partners complete deliveries faster than new delivery partners.

Let μ_1 be the mean delivery time for experienced partners and μ_2 be the mean delivery time for new partners.

Hypotheses:

- H_0: $\mu_1 \geq \mu_2$
- H_1: $\mu_1 < \mu_2$

A one-tailed independent-samples t-test, preferably Welch's t-test, can compare mean delivery times.

Analysis procedure:

1. Classify delivery partners into experienced and new groups using a clearly defined experience threshold.
2. Collect comparable delivery-time observations for both groups.
3. Calculate the mean, standard deviation, and sample size for each group.
4. Check data quality, independence, and the suitability of the test.
5. Conduct the test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.
7.

Solution: If the result is significant, the company has evidence that experienced partners have shorter average delivery times. It can use the findings to develop training programmes, improve partner allocation, and identify best practices.

The company should also consider distance, traffic, restaurant delays, and order complexity to avoid attributing every difference to experience alone.

Case Study 10: Mobile Network Performance Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help the telecom company evaluate internet speed, signal strength, call drop rate, number of users, and customer complaints across locations.

- **Mean:** Calculates average internet speed and signal strength, providing a general measure of network performance.

- **Median:** Represents the middle observation and reduces the influence of extreme network readings.
- **Standard deviation:** Measures variability in internet speed and signal strength. High variability may indicate inconsistent network quality.
- **Quartiles and IQR:** Summarize the spread of network measurements and identify unusual readings.
- **Histograms:** Show whether internet speed and signal strength are symmetrically distributed or skewed.
- **Box plots:** Compare network performance across locations and identify potential outliers.
- **Scatter plots:** Help investigate relationships between user count, signal strength, internet speed, and call drop rate.

Conclusion: The company can identify poor-performing regions, assess network congestion, and prioritize infrastructure improvements.

B. Impact and Detection of Outliers

Extremely low internet speed values can reduce the average and indicate weak coverage or congestion. Very high call drop rates may reveal network failures, overloaded cell towers, or measurement problems.

Outliers can also distort comparisons between locations and hide the typical performance experienced by most customers.

Suitable methods include:

1. **IQR method:** Identify values outside $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.
2. **Z-score method:** Flag observations with unusually large deviations from the mean when appropriate.
3. **Box plots:** Visually identify extreme readings.
4. **Domain-based validation:** Check whether measurements fall within technically possible ranges.
5. **Time and location analysis:** Compare unusual readings with network logs, tower utilization, maintenance events, and weather conditions where relevant.

The company should verify the original measurements before correcting or excluding them. Genuine poor network performance must be retained because it identifies areas requiring improvement.

Solution: Use validated data to optimize tower placement, improve bandwidth allocation, investigate equipment failures, and reduce customer complaints.

C. Confidence Intervals and Hypothesis Testing for Urban and Rural Internet Speeds

The company wants to determine whether average internet speed differs significantly between urban and rural areas.

Let μ_1 represent the mean urban internet speed and μ_2 represent the mean rural internet speed.

Hypotheses:

- $H_0: \mu_1 - \mu_2 = 0$
- $H_1: \mu_1 - \mu_2 \neq 0$

A two-tailed independent-samples t-test, preferably Welch's t-test, can compare the two means.

Procedure:

1. Collect representative internet speed measurements from urban and rural locations under comparable conditions.
2. Calculate the sample mean, standard deviation, and sample size for each group.
3. Calculate the difference between the sample means.
4. Construct a 95% confidence interval for the difference in means.
5. Conduct the two-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.

If the confidence interval excludes zero, the difference is statistically significant at the corresponding significance level.

Solution: The results can guide network investment, capacity planning, and coverage improvements. The company should also report the magnitude of the difference because a statistically significant difference may not always be practically important.

Case Study 12: Agriculture Crop Yield Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help the agriculture department understand crop yield, rainfall, fertilizer usage, soil moisture, and temperature across farms.

1. **Mean:** Calculates the average crop yield, rainfall, and fertilizer usage. It provides an overall measure of agricultural performance.
2. **Median:** Identifies the middle value and is useful when a few farms experience extremely high or low yields.
3. **Standard deviation:** Measures variability in crop yield and environmental conditions across farms.
4. **Quartiles and IQR:** Show the spread of crop yields and help identify farms with unusually low or high production.
5. **Histograms:** Display the distribution of crop yield, rainfall, and fertilizer usage.
6. **Box plots:** Compare yield distributions across regions, seasons, or fertilizer types.
7. **Scatter plots:** Help examine relationships between rainfall, fertilizer usage, soil moisture, temperature, and crop yield.

Conclusion: These analyses help identify productivity patterns, compare farming practices, and develop strategies to improve crop production.

B. Impact and Detection of Abnormal Crop Yield Values

Abnormally high or low crop yields can affect the average and lead to misleading conclusions about agricultural productivity.

Low yields may result from drought, pests, diseases, soil degradation, or insufficient fertilizer. Extremely high yields may reflect favourable growing conditions, differences in farm size, or incorrect data recording. Excessive rainfall can also reduce yield through flooding or waterlogging.

Suitable methods include:

- **IQR method:** Flag values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

- **Z-score method:** Identify values with unusually large deviations from the mean when appropriate.
- **Box plots and histograms:** Visualize extreme observations and distribution patterns.
- **Contextual validation:** Compare yields with farm size, crop variety, season, rainfall, soil conditions, and historical records.

The department should distinguish between measurement errors and genuine agricultural extremes. Incorrect data should be corrected, while genuine drought- or pest-related losses should be retained. The median can supplement the mean when extreme yields influence the results.

Solution: Validated outlier analysis can help identify farms requiring agricultural support and reveal the environmental conditions associated with crop failure.

C. Confidence Intervals and Hypothesis Testing for Fertilizer Types

The department wants to determine whether farms using organic fertilizer have significantly different crop yields from farms using chemical fertilizer.

Let μ_1 represent the mean crop yield for farms using organic fertilizer and μ_2 represent the mean yield for farms using chemical fertilizer.

Hypotheses:

- H_0: $\mu_1 - \mu_2 = 0$
- H_1: $\mu_1 - \mu_2 \neq 0$

A two-tailed independent-samples t-test, preferably Welch's t-test, can be used to compare mean yields.

Analysis procedure:

1. Collect crop yield data from representative farms using each fertilizer type.
2. Ensure yields are measured using comparable units, such as kilograms per hectare.
3. Calculate the sample mean, standard deviation, and sample size for each group.
4. Construct a 95% confidence interval for the difference in mean crop yields.
5. Perform the two-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.

A confidence interval excluding zero indicates a statistically significant difference in mean yield.

Solution: The department can use the findings to advise farmers on fertilizer selection and agricultural practices. However, soil quality, rainfall, crop variety, and farming methods may affect the results. Since farms are not necessarily randomly assigned to fertilizer types, a significant difference alone does not prove that fertilizer type caused the observed difference.

Case Study 14: Automobile Fuel Efficiency Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help the automobile company understand vehicle mileage, fuel consumption, engine capacity, vehicle weight, and maintenance cost.

- **Mean mileage:** Calculates average distance travelled per unit of fuel, allowing the company to compare overall fuel efficiency.

- **Median mileage:** Represents the middle mileage value and is less affected by extreme observations.
- **Standard deviation:** Measures variability in mileage across vehicles or driving conditions.
- **Quartiles and IQR:** Summarize the spread of mileage and identify potential extreme values.
- **Distribution analysis:** Histograms and box plots show the distribution and variation of mileage and fuel consumption.
- **Scatter plots:** Examine relationships between mileage, engine capacity, and vehicle weight.

The company should use consistent measurement units and comparable driving conditions when evaluating mileage.

Conclusion: These statistics help engineers identify efficient vehicle designs, evaluate fuel consumption, and support decisions about vehicle development.

B. Impact and Detection of Abnormal Mileage Values

Abnormal mileage values can distort the average and lead to incorrect conclusions about vehicle efficiency. Extremely high mileage may result from measurement errors or unusually favourable test conditions, while extremely low mileage may indicate mechanical faults, heavy loads, or inefficient engine performance.

Suitable methods include:

1. **IQR method:** Flag mileage values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.
2. **Z-score method:** Identify extreme standardized values when the data distribution makes the method appropriate.
3. **Box plots:** Visualize potential outliers.
4. **Engineering validation:** Compare observations with expected vehicle specifications, fuel consumption records, and test conditions.

The company should verify abnormal observations before removing them. Genuine low mileage values are important for identifying design weaknesses. Confirmed measurement errors should be corrected, and robust statistics can be used to summarize the data when genuine extremes remain.

Solution: Reliable outlier handling improves the accuracy of fuel efficiency comparisons and helps engineers identify opportunities for design improvement.

C. Confidence Intervals and Hypothesis Testing for Hybrid and Petrol Cars

The company wants to determine whether hybrid cars have significantly better mileage than petrol cars.

Let μ_1 represent the mean mileage of hybrid cars and μ_2 represent the mean mileage of petrol cars.

Hypotheses:

- $H_0: \mu_1 \leq \mu_2$
- $H_1: \mu_1 > \mu_2$

A one-tailed independent-samples t-test, preferably Welch's t-test, can compare the mean mileage of the two groups.

Procedure:

1. Collect representative mileage measurements for hybrid and petrol vehicles.
2. Use the same mileage units and comparable driving conditions.
3. Calculate the sample mean, standard deviation, and sample size for each group.
4. Estimate the difference between two sample means.
5. Construct a confidence interval for the difference in mean mileage and perform a one-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.
7.

Solution: If the result is significant, the company has evidence that hybrid cars achieve higher average mileage than petrol cars under the studied conditions. It can use the findings to guide vehicle development and communicate fuel efficiency accurately.
The company should also consider vehicle weight, engine design, driving conditions, and test procedures to ensure a fair comparison.

Case Study 16: Airline Passenger Satisfaction Analysis

A. Descriptive Statistics for Delays and Satisfaction

Descriptive statistics help the airline understand flight delay time, baggage waiting time, ticket price, seat comfort ratings, and overall passenger satisfaction.

1. **Mean:** Calculates the average flight delay, baggage waiting time, and satisfaction score.
2. **Median:** Identifies the middle delay or satisfaction value and is useful when a small number of flights experience severe delays.
3. **Standard deviation:** Measures the variability of delays and satisfaction ratings.
4. **Quartiles and IQR:** Show the spread of waiting times and help identify unusually long delays.
5. **Histograms:** Display the distributions of flight delays and satisfaction scores.
6. **Box plots:** Identify potential outliers and compare satisfaction between different flight groups.
7. **Relationship analysis:** Scatter plots can investigate whether longer delays are associated with lower satisfaction.

Conclusion: These statistics help the airline identify service weaknesses, improve baggage handling, and develop measures to increase passenger satisfaction.

B. Impact and Detection of Unusually Long Flight Delays

Unusually long flight delays can increase the mean delay time and make overall service performance appear worse. Such delays may also be associated with lower satisfaction scores, missed connections, and increased complaints.
Possible causes include adverse weather, technical problems, air traffic congestion, staffing shortages, and operational disruptions.
Suitable detection methods include:

- **IQR method:** Flag delays above $Q3 + 1.5 \times IQR$ as potential outliers.
- **Z-score method:** Identify unusually high delay times when appropriate.

- **Box plots:** Visually identify extreme delay observations.
- **Operational validation:** Compare flagged delays with flight logs, weather reports, and airport operational records.

The airline should investigate each flagged delay rather than automatically removing it. Genuine disruptions are important for evaluating service quality. Incorrect records should be corrected, while robust statistics can help summarize the typical delay time.

Solution: The airline should use the findings to improve contingency planning, aircraft scheduling, passenger communication, and operational reliability.

C. Hypothesis Testing for On-Time and Delayed Flights

The airline wants to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights.

Let μ_1 represent the mean satisfaction rating for passengers on on-time flights and μ_2 represent the mean rating for passengers on delayed flights.

Hypotheses:

- H_0: $\mu_1 \leq \mu_2$
- H_1: $\mu_1 > \mu_2$

A one-tailed independent-samples t-test, preferably Welch's t-test, can compare the mean satisfaction ratings.

Testing procedure:

1. Classify flights as on-time or delayed using a clearly defined operational threshold.
2. Collect passenger satisfaction ratings for both groups.
3. Calculate the mean, standard deviation, and sample size for each group.
4. Check data quality and the assumptions of the selected test.
5. Conduct the one-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.
7.

Solution: A significant result provides evidence that passengers on on-time flights report higher average satisfaction. The airline can use this evidence to prioritize punctuality improvements and improve passenger communication during disruptions.

Other factors, such as seat comfort, baggage handling, ticket price, and flight duration, should also be considered when interpreting satisfaction ratings.

Case Study 18: Hotel Booking and Customer Review Analysis

A. Descriptive Statistics for Room Price, Length of Stay, and Ratings

Descriptive statistics help the hotel chain understand room prices, length of stay, customer ratings, booking cancellations, and service feedback.

- **Mean room price:** Calculates the average room price and supports pricing decisions.
- **Median room price:** Represents the middle price and reduces the influence of unusually expensive bookings.
- **Mean and median length of stay:** Summarize the duration of customer visits and help forecast room demand.
- **Standard deviation:** Measures variability in room prices and length of stay.

- **Quartiles and IQR:** Show the spread of prices and identify unusually high or low values.
- **Rating distribution:** Summarizes customer satisfaction and reveals whether most ratings are high, moderate, or low.
- **Histograms and box plots:** Show the distributions of room prices, stay durations, and ratings.

The hotel should compare room prices within similar room categories, seasons, and booking conditions to avoid misleading conclusions.

Conclusion: These statistics support revenue management, demand forecasting, service improvement, and customer satisfaction analysis.

B. Impact and Detection of Abnormal Room Prices or Customer Ratings

Unusually high room prices can increase the mean and distort the hotel's understanding of typical pricing. Extremely low customer ratings can reduce the average satisfaction score and may indicate service failures or dissatisfied guests.

Potential causes of extreme room prices include holiday seasons, premium room categories, special events, and pricing errors. Low ratings may reflect poor cleanliness, inadequate service, or incorrect review records.

Suitable methods include:

1. **IQR method:** Identify prices or ratings outside the expected range based on Q1, Q3, and IQR.
2. **Z-score method:** Flag extreme numerical observations when appropriate.
3. **Box plots:** Visualize potential outliers.
4. **Contextual validation:** Compare prices with room categories, dates, occupancy levels, and booking conditions.
5. **Rating validation:** Check whether reviews are genuine and recorded correctly.

The hotel should not remove genuine high prices or low ratings simply because they are extreme. Invalid transactions and data errors should be corrected, while valid observations should be retained and investigated.

Solution: The hotel can use verified outliers to identify pricing anomalies, improve service quality, and develop better revenue management strategies.

C. Hypothesis Testing for Length of Stay and Customer Ratings

The hotel wants to determine whether customers who stay more than three days give significantly different ratings from customers who stay for one or two days.

Let μ_1 represent the mean customer rating for stays longer than three days and μ_2 represent the mean rating for stays of one or two days.

Hypotheses:

- H_0: $\mu_1 = \mu_2$
- H_1: $\mu_1 \neq \mu_2$

A two-tailed independent-samples t-test, preferably Welch's t-test, can compare the mean ratings. Since ratings are often ordinal and may have a restricted range, an ordinal or non-parametric alternative may be appropriate if the assumptions of the t-test are unsuitable.

Analysis procedure:

1. Divide customers into the two specified length-of-stay groups.
2. Collect valid customer ratings for each group.
3. Calculate the mean, standard deviation, and sample size for each group.
4. Check data quality and the suitability of the statistical test.
5. Conduct the two-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.
7.

Solution: A significant result indicates evidence of a difference in average ratings between the two groups. The hotel should investigate whether longer stays are associated with better or worse service experiences and use the findings to improve guest services.
The results should be interpreted carefully because room type, customer expectations, travel purpose, and price may also influence ratings.

Case Study 20: Online Examination Performance Analysis

A. Descriptive Statistics and Distribution Analysis

Descriptive statistics help the educational institution understand examination scores, completion time, number of attempts, internet interruptions, and student feedback.

1. **Mean examination score:** Calculates the average student score and summarizes overall performance.
2. **Median score:** Identifies the middle examination score and is less affected by extreme marks.
3. **Standard deviation:** Measures variation in scores and shows whether student performance is consistent.
4. **Quartiles and IQR:** Divide scores into four groups and identify the spread of examination performance.
5. **Completion-time analysis:** Mean, median, and standard deviation summarize how long students take to complete examinations.
6. **Attempt analysis:** The distribution of attempts helps identify whether students generally complete examinations on their first attempt or require repeated attempts.
7. **Distribution analysis:** Histograms and box plots reveal skewness and potential outliers in scores and completion times.
8. **Relationship analysis:** Scatter plots can examine the relationship between completion time, number of attempts, internet interruptions, and examination scores.

Conclusion: These analyses help the institution evaluate examination difficulty, identify technical issues, and improve the reliability of the online assessment system.

B. Impact and Detection of Abnormal Completion Times or Attempts

Unusually short completion times may indicate that a student finished quickly, submitted incomplete answers, experienced a recording error, or potentially engaged in irregular examination behaviour. However, short completion time alone does not prove misconduct.
An extremely high number of attempts may reflect internet interruptions, repeated login failures, technical problems, or confusion about the examination process.
Suitable methods include:

- **IQR method:** Identify completion times or attempt counts outside $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.
- **Z-score method:** Flag unusually extreme values where the data distribution makes the method appropriate.
- **Box plots and histograms:** Identify potential outliers and skewed distributions.
- **System-log analysis:** Verify submission timestamps, login records, interruption counts, and attempt histories.
- **Domain-based checks:** Compare completion times with the examination duration and attempt counts with the permitted maximum.

The institution should investigate unusual records before taking action. Verified technical errors should be corrected, while genuine unusual performance should be retained. Potential misconduct should be assessed using multiple sources of evidence and established institutional procedures.

Solution: A systematic outlier-handling process improves the accuracy of performance analysis and helps distinguish genuine student behaviour from technical problems.

C. Hypothesis Testing for Internet Interruptions and Examination Scores

The institution wants to determine whether students with fewer internet interruptions score significantly higher than students with frequent interruptions.

Let μ_1 represent the mean examination score of students with fewer interruptions and μ_2 represent the mean score of students with frequent interruptions.

The institution should define the interruption threshold before analysing the data.

Hypotheses:

- H_0: $\mu_1 \leq \mu_2$
- H_1: $\mu_1 > \mu_2$

A one-tailed independent-samples t-test, preferably Welch's t-test, can compare the mean scores of the two groups.

Testing procedure:

1. Collect examination scores and internet interruption counts for all eligible students.
2. Define the two groups using a clear and consistent threshold.
3. Calculate the mean, standard deviation, and sample size for each group.
4. Check data quality, independence, and the suitability of the selected test.
5. Conduct the one-tailed test at a 5% significance level.
6. Reject H_0 if the p-value is below 0.05.
7.

Solution: If the result is statistically significant, the institution has evidence that students with fewer internet interruptions achieve higher average examination scores. It can use this information to improve server capacity, strengthen network reliability, provide technical support, and introduce suitable examination recovery procedures.

The institution should also consider prior academic ability, device quality, examination difficulty, and other relevant factors. A significant difference establishes an association but does not, by itself, prove that internet interruptions caused lower scores.